

Abstract:

This project focuses on the design and implementation of a D Flip-Flop with Synchronous reset. The objective was to develop a medium-scale integration chip following a structured top-down design hierarchy, from system-level functional analysis to complete chip layout. The D Flip-Flop with Synchronous Reset was designed to store a single bit of data, with the reset functionality synchronized to the clock signal to ensure stable operation. The design process included functional and architectural analysis, top-level layout placement, routing, I/O pad placement, and post-layout simulation to verify functionality and eliminate Design Rule Check (DRC) errors. The project adhered to the specified guidelines, ensuring proper documentation of the chip's specifications, state diagrams, and verification at each stage. The final design successfully met the functional requirements, demonstrating reliable data storage and reset operation, validated through simulation and layout checks.

Introduction:

Digital circuits constitute the foundation of contemporary computing systems, undertaking vital functions such as data storage, processing, and control. Central to these circuits are flip flops, which are critical components that store and synchronize binary data within sequential logic circuits, memory units, and state machines. Historically, flip flops have been essential in their architecture, primarily providing fundamental functionalities like data storage and clocked output. Nevertheless, as digital systems evolve in complexity and sophistication, there emerges an urgent requirement for more advanced features to improve reliability, flexibility, and efficiency. One notable advanced feature is the incorporation of synchronous reset capability in flip flops. This synchronous reset ensures that the stored data can be cleared or reset in alignment with a clock signal, typically occurring on designated clock cycles. This feature is vital in scenarios where precise control over data retention and reset operations is critical. By aligning reset actions with the clock, it mitigates unintended resets and safeguards data integrity during crucial operations. The addition of synchronous reset significantly enhances the overall reliability and stability of digital circuits, especially in contexts where accurate timing and data consistency are imperative. It enables controlled resetting of flip flop states, which is essential in applications ranging from microprocessor design to communication protocols and digital signal processing.

Objectives:

The project covered:

- Behavioral and architectural planning
- Gate-level and transistor-level schematic design
- Layout creation using standard design rules
- DRC (Design Rule Check) and LVS (Layout vs. Schematic) validation
- Post-layout simulation
- GDSII file generation for tape-out readiness

Tools used: Cadence Virtuoso.

System Design:

Working of D flip flop:

D flip flop consist of a single input D and two outputs (Q and Q'). The basic working of D Flip Flop is as follows:
When the clock signal is low, the flip flop holds its current state and ignores the D input.
When the clock signal is high, the flip flop samples and stores D input.

The value that was previously fed into the D input is reflected at the flip flop's Q output.

If D = 0 then Q will be 0.

If D = 1 then Q will be 1.

The Q' output of the flip flop is complemented by the Q output.

If Q = 0 then Q' will be 1.

If Q = 1 then Q' will be 0.

Characteristic table of D flip flop:

D flip flop consist of a single input D and two outputs (Q and Q'). The basic working of D Flip Flop is as follows:

When the clock signal is low, the flip flop holds its current state and ignores the D input.

When the clock signal is high, the flip flop samples and stores D input.

The value that was previously fed into the D input is reflected at the flip flop's Q output.

If D = 0 then Q will be 0.

If D = 1 then Q will be 1.

The Q' output of the flip flop is complemented by the Q output.

If Q = 0 then Q' will be 1.

If Q = 1 then Q' will be 0.

Table 01: Characteristic table of D flip-flop

D	Q(Current)	Q(n+1) (Next)
0	0	0
0	1	0
1	0	1
1	1	1

D is the input, and Q is current state, Q(n+1) is the next state outputs.

Q(n+1) will always be zero when D is 0, irrespective of current state of flip flop.

When the input of the flip flop is 1, next state of flip flop will always be 1, regardless of the current state of flip flop.

Characteristic equation of D flipflop:

The characteristics equation of D flip flop consists of a Boolean expression that explains the relationship between the input and output of the flip flop. The characteristic equation for a D flip flop is:

$$Q(n + 1) = D(n)$$

Q(n+1) represents the output of flip flop at the next clock cycle.

D(n) is the input to the flip flop at the current clock cycle, and n represents the current clock cycle.

This characteristic equation of D flip flop states that the output of the flip flop at the next clock cycle will be equal to the input at the current clock cycle.

Block diagram of D flipflop with synchronous reset:

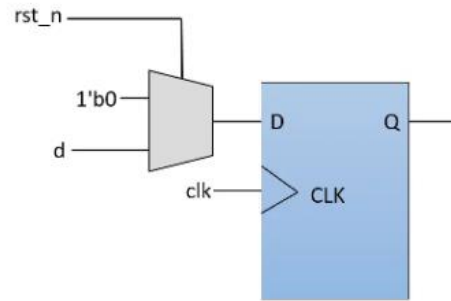


Figure 01: Block diagram of D flipflop with synchronous reset

Truth Table:

Table 02: Truth table of D Flip-flop

Input			Output	
D	Reset	Clock	Q	Q'
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	1	0
1	1	0	0	1
1	1	1	0	1

System design parameters:

To design the schematic of the D flip flop-

For the ground -

- voltage is 0 volt

For the Vdd-

- voltage is 1.2 volt.

For the Reset input-

- Voltage is 1.2 Volt.
- Period is 100 ms.
- Pulse Width 50 ms.

For the data input-

- Voltage is 1.2 Volt.
- Period is 40ms.
- Pulse Width 20 ms.

For the Clock input-

- Voltage is 1.2 Volt.
- Period is 20 ms.
- Pulse Width 10 ms

Gate level schematic design and Symbol creation:

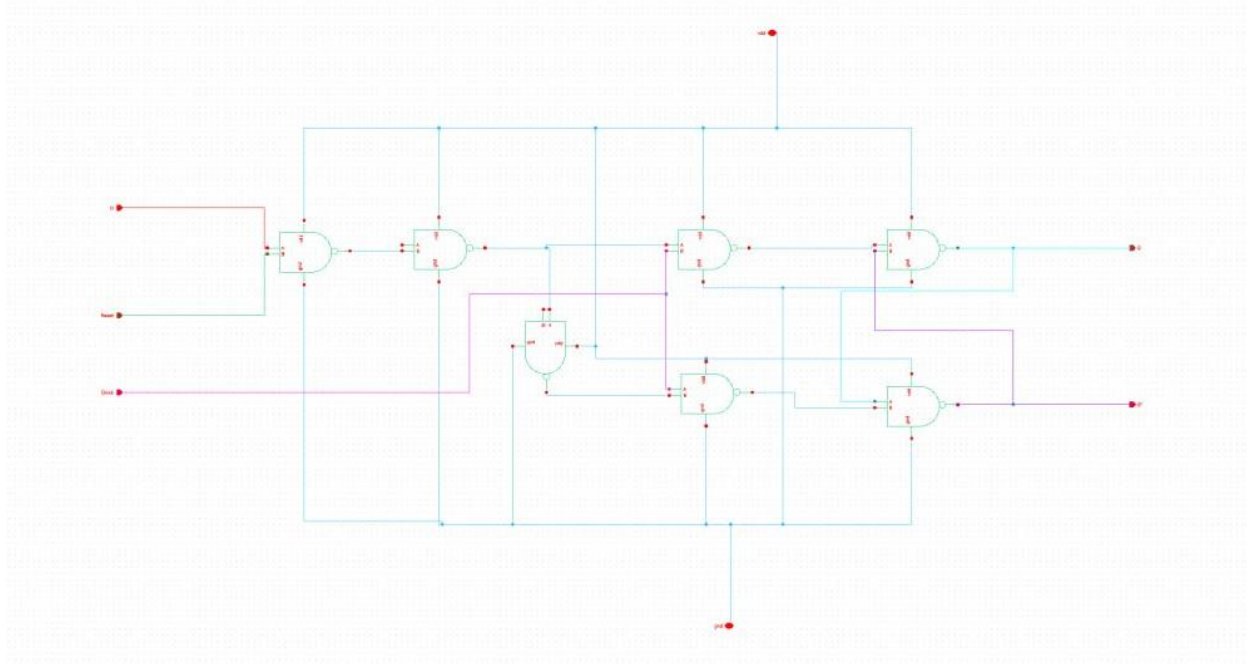


Figure 02: Gate level design of flip the D flipflop with synchronous reset

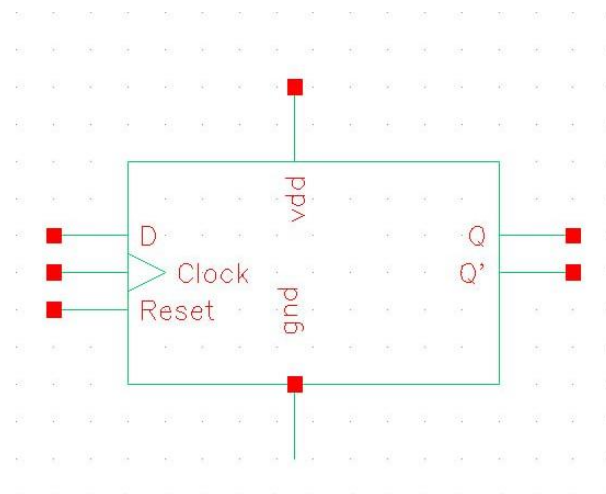


Figure 03: Symbol creation of D flip-flop with synchronous reset

Output of the schematic:

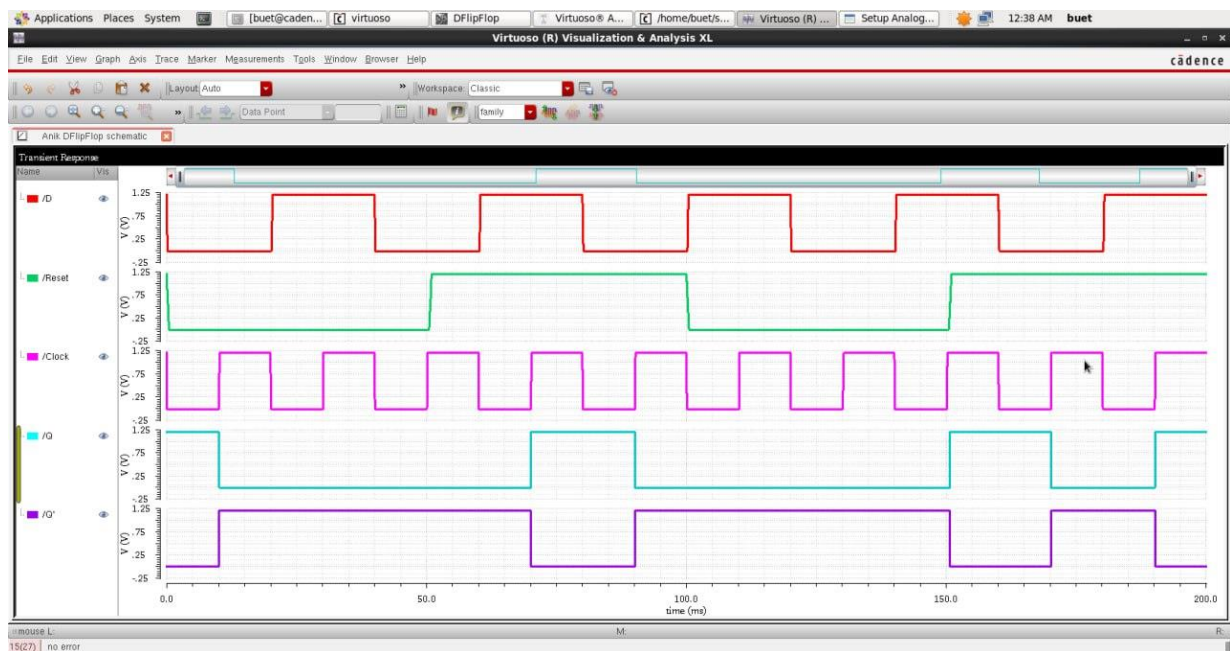


Figure 04: Output with all combined signal



Figure 05: Transient response with reset value high



Figure 06: Transient response with reset value low

Transistor level design and Layout design:

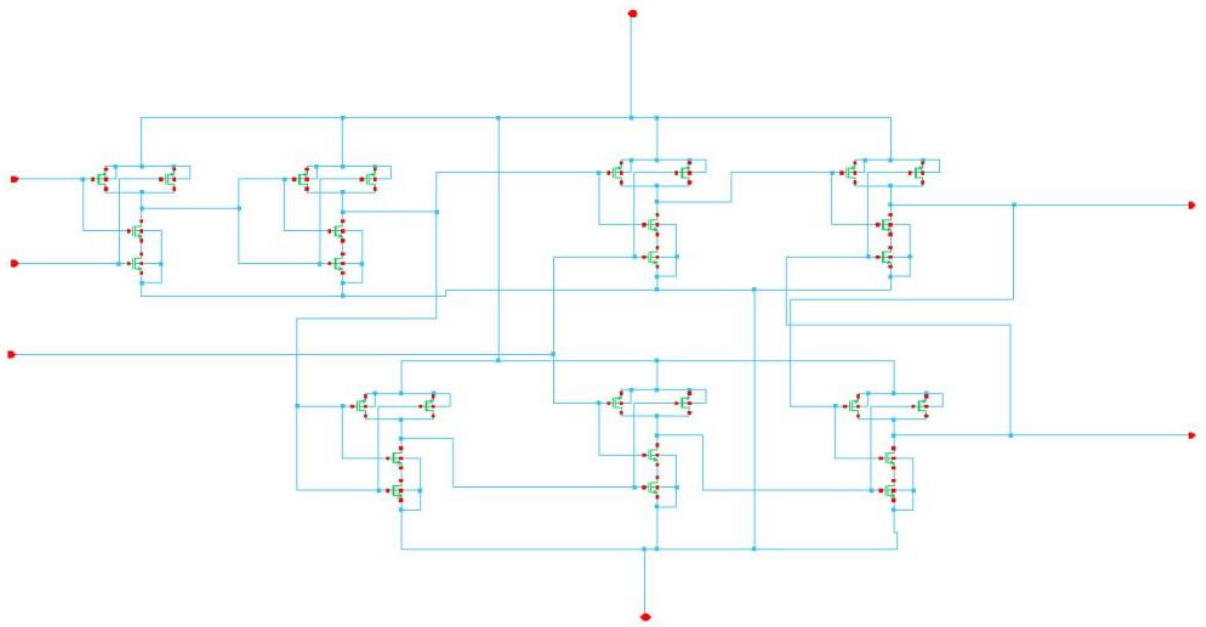


Figure 07: Transistor level design of D flip flop with synchronous reset

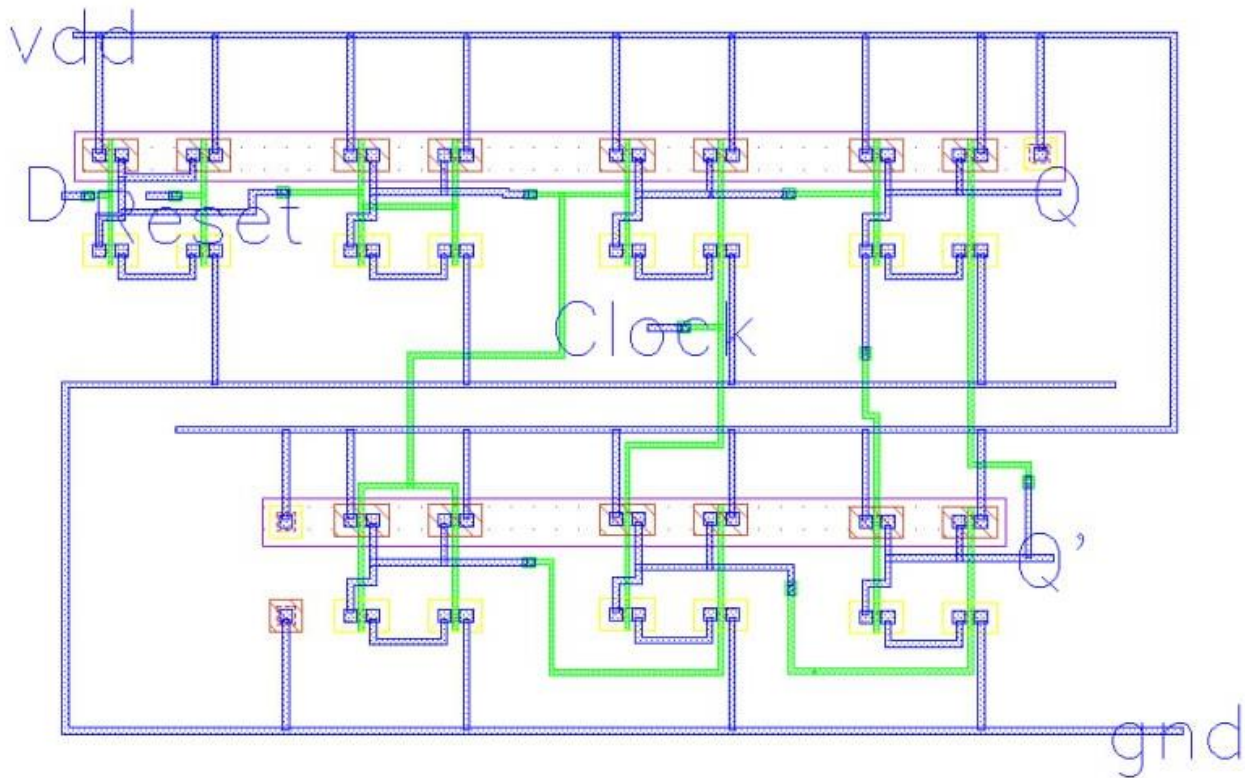


Figure 08 : Top level Layout design

DRC and LVS check:

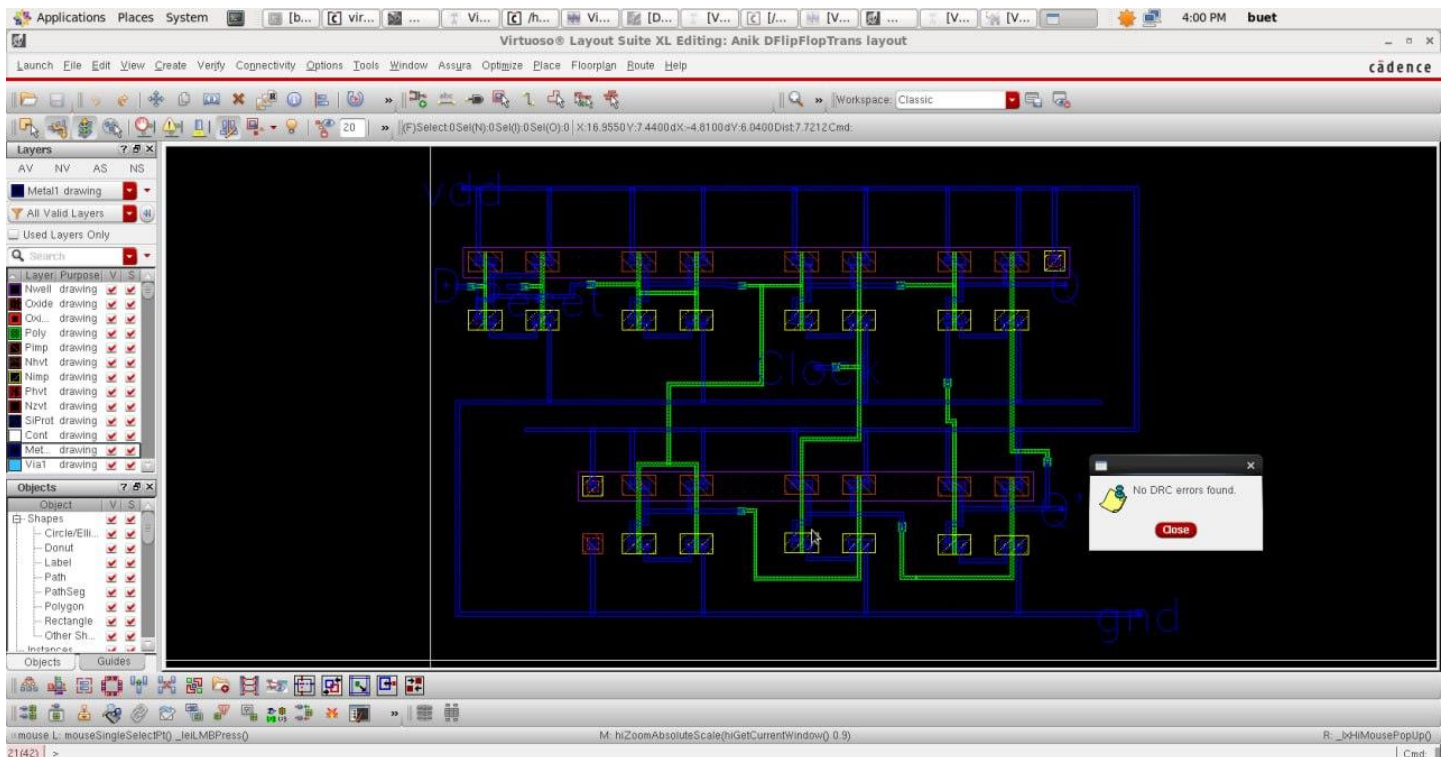


Figure 09: DRC Run successful

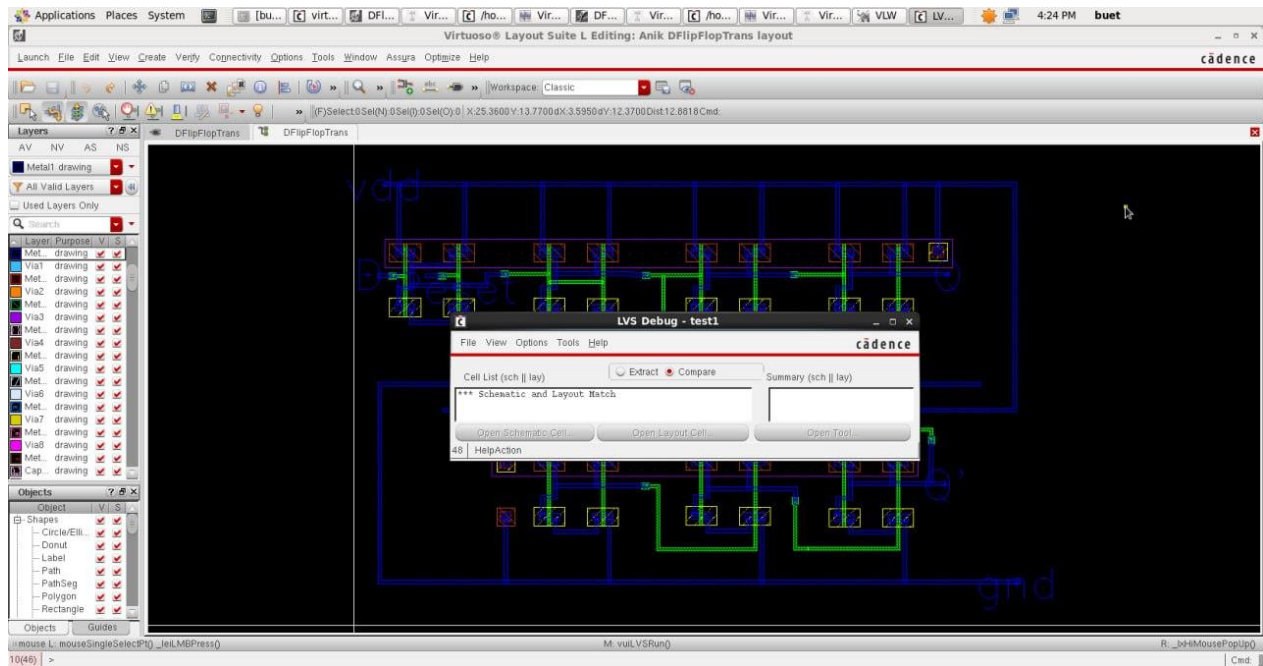


Figure 10: LVS check successful

Chip layout with I/O pad placement, DRC run and Post Layout simulation:

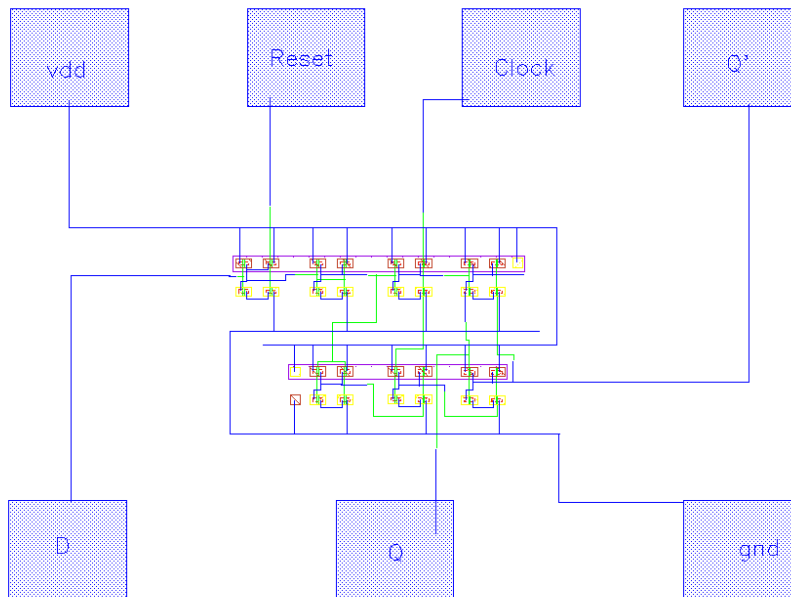


Figure 11: Chip layout with I/O pad placement

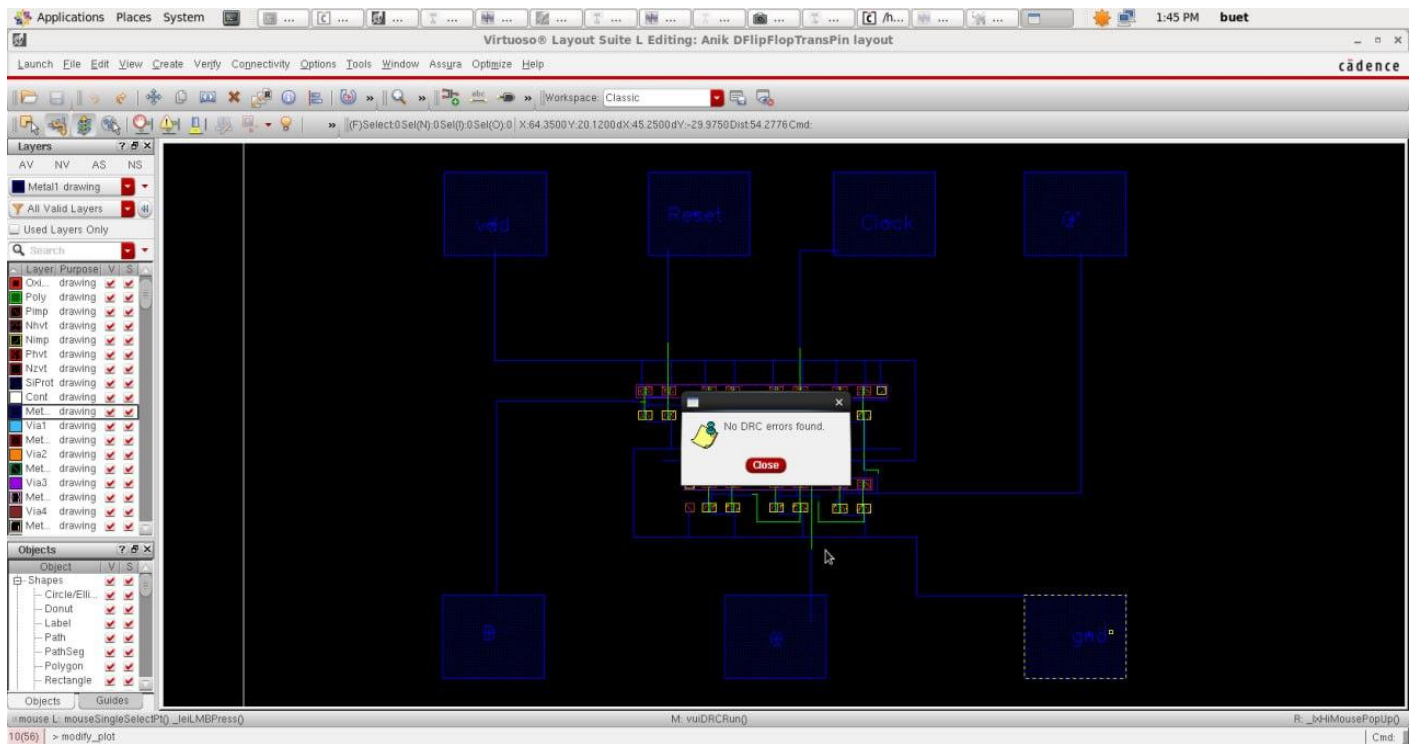


Figure 12: DRC check after I/O pad placement

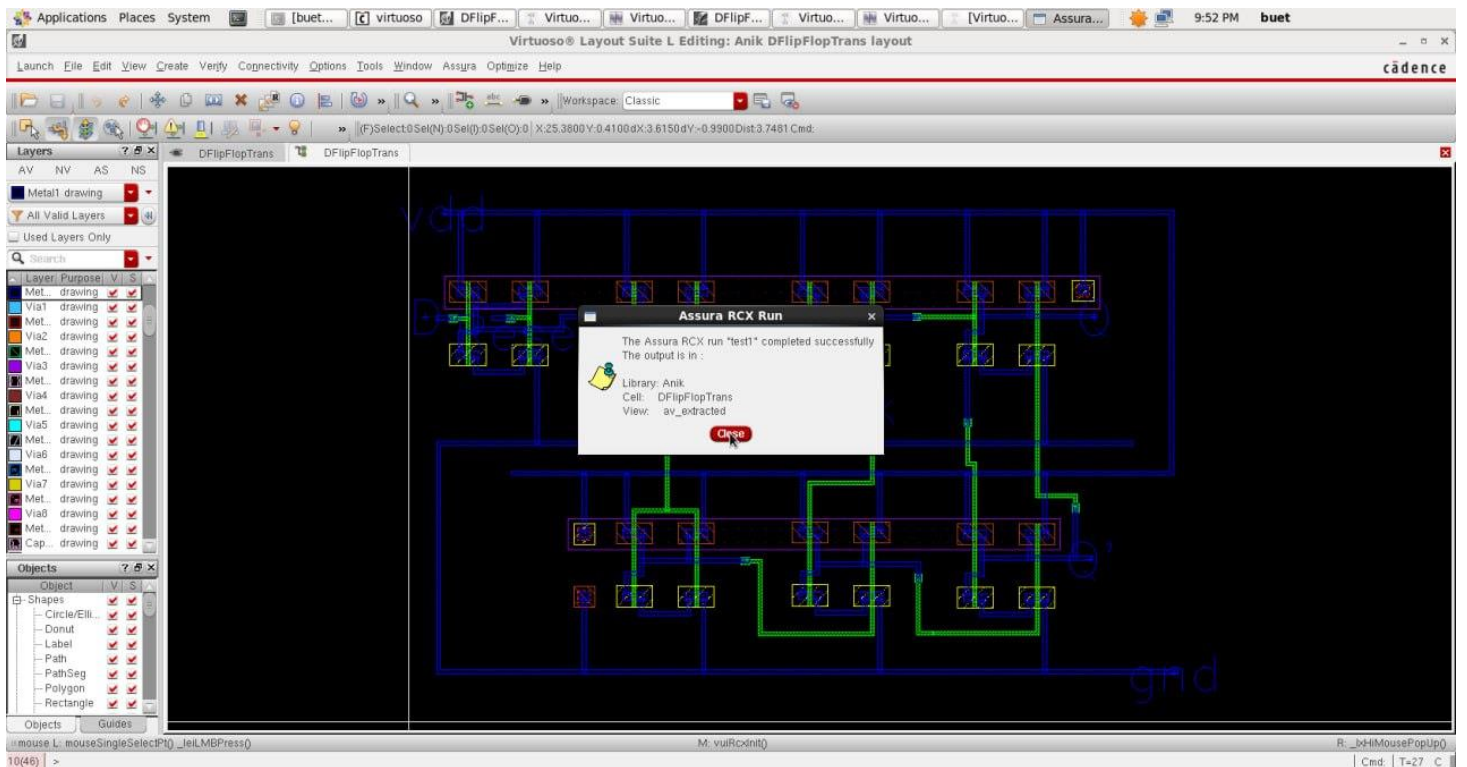


Figure 13: RCX run successful

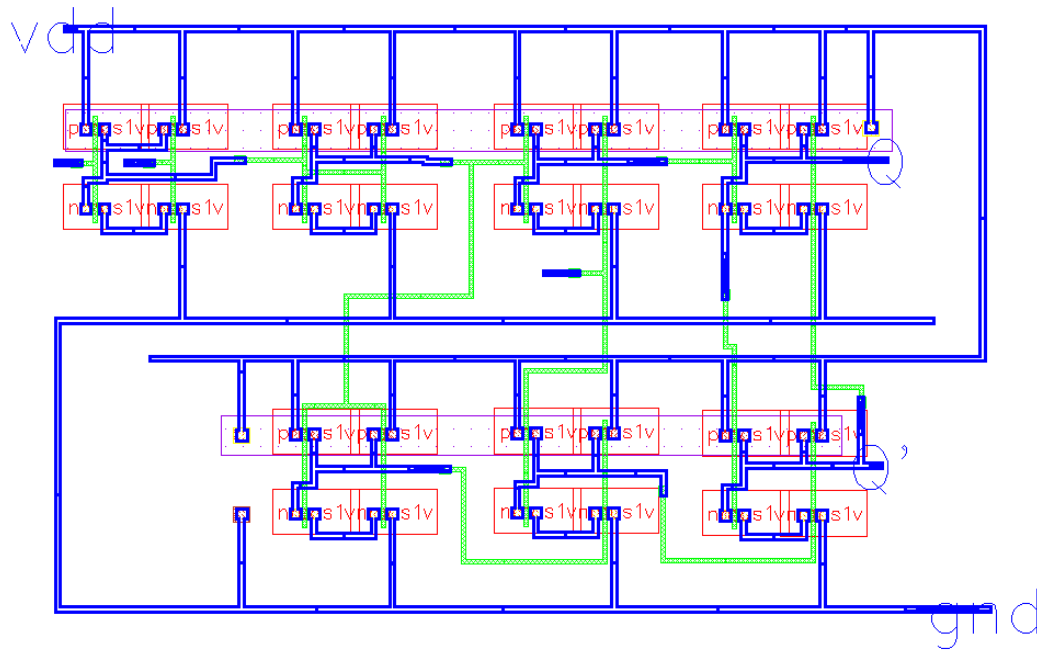


Figure 14: AV extracted view of layout

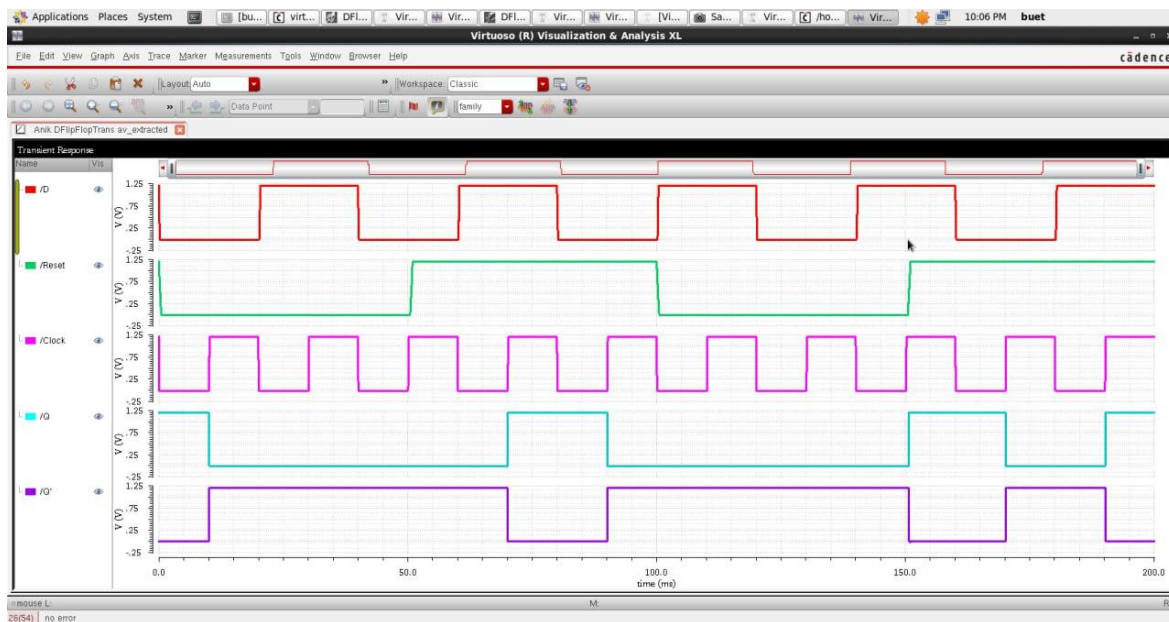


Figure 15: Output after post layout simulation

Conclusion:

The successful completion of the D Flip-Flop with Synchronous Reset project demonstrates the application of structured VLSI design principles to achieve a functional medium scale integration chip. Through systematic progression from system design to final layout, the project met all specified objectives, including functional accuracy, architectural optimization, and error-free layout implementation. The design process highlighted the importance of thorough verification at each hierarchy level, ensuring the D Flip-Flop operated reliably with synchronous reset functionality. The project also fostered effective teamwork and adherence to design constraints, resulting in a robust chip layout with properly placed I/O pads and cleared DRC errors. This work provides a solid foundation for further exploration of sequential logic design in VLSI systems, with potential applications in digital circuits requiring synchronized data storage and reset capabilities.

